

Exponentiation Digest

Exponents

$$x^0 = 1$$

$$x^1 = x$$

$$x^2 = x \cdot x$$

$$x^3 = x \cdot x \cdot x$$

$$\dots$$
$$(b^x)^y = (b^y)^x = b^{xy}$$

$$b^{-x} = \frac{1}{b^x}$$

$$a^x \cdot b^x = (ab)^x$$

$$b^x \cdot b^y = b^{x+y}$$

$$b^x / b^y = b^{x-y}$$

Roots

$$b^x = y \Leftrightarrow b = \sqrt[x]{y}$$

$\sqrt[0]{b}$ is undefined

$$\sqrt[x]{b^x} = \left(\sqrt[x]{b}\right)^x = \begin{cases} b & \text{if } x \text{ is odd} \\ |b| & \text{if } x \text{ is even} \end{cases}$$

$$\sqrt[x]{b} = b^{\frac{1}{x}}$$

$$\sqrt{x} = \sqrt[2]{x}$$

Logarithms

$$b^x = y \Leftrightarrow \log_b(y) = x$$

$$b^1 = b \Leftrightarrow \log_b(b) = 1$$

$$b^0 = 1 \Leftrightarrow \log_b(1) = 0$$

$$\log_b \Rightarrow b > 0, b \neq 1$$

$$\log_b(z) \text{ is undefined if } z \leq 0$$

$$\log_b(b^x) = b^{\log_b(x)} = x$$

$$\log_b(x^y) = y \cdot \log_b(x)$$

$$\log_b(x \cdot y) = \log(x) + \log(y)$$

$$\log_b(x/y) = \log(x) - \log(y)$$

$$\frac{\log_a(x)}{\log_a(b)} = \log_b(x)$$

$$\frac{1}{\log_x(b)} = \log_b(x)$$

$$\log_b(a) \cdot \log_a(x) = \log_b(x)$$

$$\log(x) = \log_{10}(x)$$

$$\ln(x) = \log_e(x)$$

TODO

- graphs
 - < 1 and > 1 base variants
- exponential function e^x
- $\sqrt{-1}$
- calculus?